MA-310 (Calculus III)

Worcester State University

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Contents

Chapter 1

Vectors and the Geometry of Space

Vector calculus is the branch of mathematics that deals with vector fields and their derivatives. It is essential for understanding the behavior of physical systems in three-dimensional space.

In this chapter, we will explore the fundamental concepts of vector calculus, including gradient, divergence, and curl.

1.1 Vectors in Two Dimensions

A vector is a quantity that has both magnitude and direction. In two dimensions, vectors can be represented as arrows in a plane, where the length of the arrow corresponds to the magnitude of the vector, and the direction of the arrow indicates the direction of the vector.

Definition 1.1 A velocity or force has both magnitude and direction and is often represented by a **directed line segment** or a **vector**. We denote the vector with **initial point** P and **terminal point** Q as $\vec{PQ}$. The direction of the vector indicates arrowhead at Q. The **magnitude** of $\vec{PQ}$ is the length of the segment PQ and is denoted by $\|\vec{PQ}\|$. Vectors that have the same magnitude and direction are called **equivalent**. ◆

A vector may be translated from one location to another, provided neither the magnitude nor the direction is changed. If $\vec{PS} = \vec{PQ} + \vec{PR}$ and $\vec{PQ}$ and $\vec{PR}$ are two forces acting at P, then $\vec{PS}$ is the **resultant force** (sum).

If c is a scalar and $\vec{v}$ is a vector, then $c\vec{v}$ (a **scalar multiple**) is defined as a vector whose magnitude is $|c|$ times the magnitude $\|\vec{v}\|$ of $\vec{v}$ with the same direction ($c > 0$) or the opposite direction ($c < 0$) of $\vec{v}$.

A vector with the initial point at the origin is called the **position vector**. We use the symbol $\langle a_1, a_2\rangle$ for an ordered pair that represents the vector and denote it by a lowercase letter $\vec{v} = \langle a_1, a_2\rangle$.

Two vectors $\langle a_1, a_2\rangle$ and $\langle b_1, b_2\rangle$ are **equal** $\iff$ $a_1 = b_1$ and $a_2 = b_2$.

Example 1.2 Sketch the position vectors for:

1 $\vec{a} = \langle -3, 2\rangle$

2 $\vec{b} = \langle 0, -2\rangle$

3 $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5}\right\rangle$

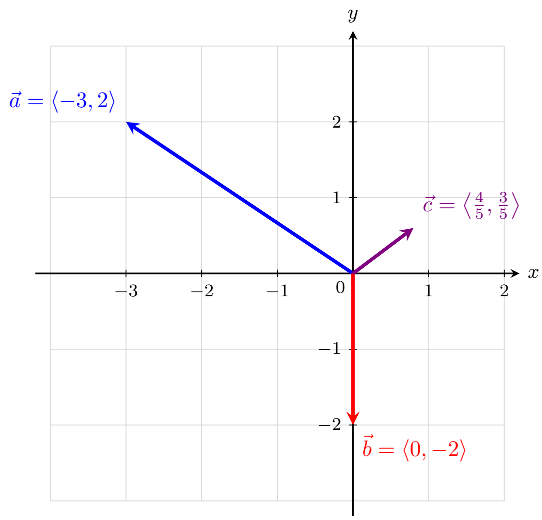

Figure 1.3 The position vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ drawn in the Cartesian plane.

Solution. To sketch a position vector, always place its initial point at the origin $(0, 0)$. The terminal point will match the components of the vector:

1 For $\vec{a} = \langle -3, 2 \rangle$, draw an arrow starting at $(0, 0)$ and ending at the point $(-3, 2)$.

2 For $\vec{b} = \langle 0, -2 \rangle$, draw an arrow starting at $(0, 0)$ and pointing straight down to end at the point $(0, -2)$.

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$, draw an arrow starting at $(0, 0)$ and ending at the point $(0.8, 0.6)$ in the first quadrant.

▲

Definition 1.4 The **magnitude** (or length) $\|\vec{a}\|$ of the vector $\vec{a} = \langle a_1, a_2 \rangle$ is

$$\|\vec{a}\| = \|\langle a_1, a_2 \rangle\| = \sqrt{a_1^2 + a_2^2}.$$

◆

Example 1.5 Find the magnitude of each vector:

1 $\vec{a} = \langle -3, 2 \rangle$

2 $\vec{b} = \langle 0, -2 \rangle$

3 $\vec{c} = \left\langle \dfrac{4}{5}, \dfrac{3}{5} \right\rangle$

Solution. We apply the magnitude formula $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$ to each vector:

1 For $\vec{a} = \langle -3, 2 \rangle$:

$$\|\vec{a}\| = \sqrt{(-3)^2 + 2^2} = \sqrt{9 + 4} = \sqrt{13}$$

2 For $\vec{b} = \langle 0, -2 \rangle$:

$$\|\vec{b}\| = \sqrt{0^2 + (-2)^2} = \sqrt{0 + 4} = \sqrt{4} = 2$$

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$:

$$\|\vec{c}\| = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25} + \frac{9}{25}} = \sqrt{\frac{25}{25}} = 1$$

Because its magnitude is 1, $\vec{c}$ is a unit vector.

▲

Fact 1.6 Vector Operations. ***Addition of vectors***

$$\langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$$

Scalar multiple of a vector

$$c\langle a_1, a_2 \rangle = \langle ca_1, ca_2 \rangle$$

Example 1.7 Let $\vec{a} = \langle -1, 3 \rangle$ and $\vec{b} = \langle 4, 3 \rangle$.

1 Find $\vec{a} + \vec{b}$ and $-3\vec{a} + 2\vec{b}$.

2 Sketch vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

Solution.

1 To find $\vec{a} + \vec{b}$, add the corresponding components:

$$\vec{a} + \vec{b} = \langle -1 + 4, 3 + 3 \rangle = \langle 3, 6 \rangle$$

To find $-3\vec{a} + 2\vec{b}$, apply scalar multiplication first, then add:

$$-3\vec{a} = -3\langle -1, 3 \rangle = \langle 3, -9 \rangle$$

$$2\vec{b} = 2\langle 4, 3 \rangle = \langle 8, 6 \rangle$$

$$-3\vec{a} + 2\vec{b} = \langle 3 + 8, -9 + 6 \rangle = \langle 11, -3 \rangle$$

2 First, calculate the scalar multiple: $-\frac{3}{2}\vec{b} = -\frac{3}{2}\langle 4, 3 \rangle = \left\langle -6, -\frac{9}{2} \right\rangle = \langle -6, -4.5 \rangle$.

Below is the sketch of the two position vectors on the coordinate plane:

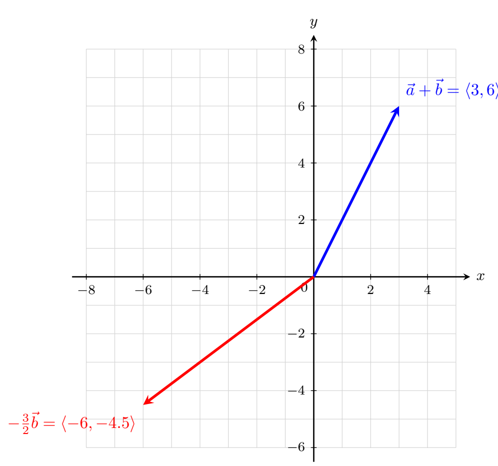

Figure 1.8 The vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

▲

Definition 1.9 The **zero vector 0** is defined as $\vec{0} = \langle 0, 0 \rangle$, and the **negative -a** of a vector $\vec{a} = \langle a_1, a_2 \rangle$ is defined as:

$$-\vec{a} = -\langle a_1, a_2 \rangle = \langle -a_1, -a_2 \rangle$$

◆

Theorem 1.10 Properties of Vectors. *Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be vectors, and let c and d be scalars. Then:*

1 $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, *and* $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$

2 $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$

3 $\vec{a} + \vec{0} = \vec{a}$, *and* $\vec{a} + (-\vec{a}) = \vec{0}$

4 $(c + d)\vec{a} = c\vec{a} + d\vec{a}$, *and* $(cd)\vec{a} = c(d\vec{a}) = d(c\vec{a})$

5 $1\vec{a} = \vec{a}$, *and* $0\vec{a} = \vec{0} = c\vec{0}$

Definition 1.11 Let $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$. The **difference a-b** is:

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2 \rangle$$

◆

Example 1.12 If $\vec{a} = \langle 5, -3 \rangle$ and $\vec{b} = \langle -2, 3 \rangle$, find:

1 $\vec{a} - \vec{b}$

2 $3\vec{a} - 2\vec{b}$

Solution.

1 To find $\vec{a} - \vec{b}$, subtract the components of $\vec{b}$ from the components of $\vec{a}$:

$$\vec{a} - \vec{b} = \langle 5 - (-2), -3 - 3 \rangle = \langle 5 + 2, -6 \rangle = \langle 7, -6 \rangle$$

2 To find $3\vec{a} - 2\vec{b}$, perform the scalar multiplications first, then subtract:

$$3\vec{a} = 3\langle 5, -3 \rangle = \langle 15, -9 \rangle$$

$$2\vec{b} = 2\langle -2, 3 \rangle = \langle -4, 6 \rangle$$

Now subtract the resulting components:

$$3\vec{a} - 2\vec{b} = \langle 15 - (-4), -9 - 6 \rangle = \langle 15 + 4, -15 \rangle = \langle 19, -15 \rangle$$

▲

Theorem 1.13 Vector from Two Points. *If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are any two points, the vector $\vec{a}$ in V_2 that corresponds to the directed line segment $\vec{P_1P_2}$ is:*

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1 \rangle$$

Example 1.14 Given points $P(3, 2)$ and $Q(-1, 5)$:

1 Sketch $\vec{PQ}$ and $\vec{QP}$.

2 Find vectors $\vec{a}$ and $\vec{b}$ in V_2 that correspond to $\vec{PQ}$ and $\vec{QP}$.

3 Sketch the position vectors for $\vec{a}$ and $\vec{b}$.

Solution.

1 To sketch the directed line segments, draw an arrow starting at point $P(3,2)$ and ending at point $Q(-1,5)$ for $\vec{PQ}$. For $\vec{QP}$, reverse the direction so it starts at $Q(-1,5)$ and ends at $P(3,2)$. (See the left graph below).

2 To find the vector components, subtract the initial coordinates from the terminal coordinates ("terminal minus initial"):

For $\vec{a}$ corresponding to $\vec{PQ}$:

$$\vec{a} = \langle -1-3, 5-2 \rangle = \langle -4, 3 \rangle$$

For $\vec{b}$ corresponding to $\vec{QP}$:

$$\vec{b} = \langle 3-(-1), 2-5 \rangle = \langle 4, -3 \rangle$$

3 To sketch the position vectors $\vec{a}$ and $\vec{b}$, shift them so their initial points sit exactly at the origin $(0,0)$. The terminal points will be $(-4,3)$ and $(4,-3)$, respectively. Notice that they are perfectly parallel to and have the same lengths as the original segments (See the right graph below).

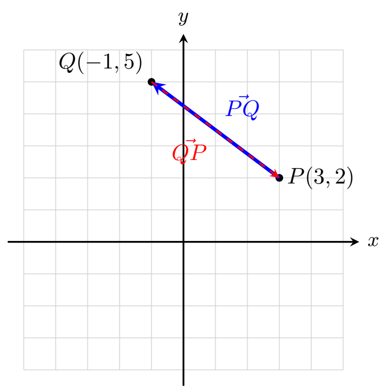

Figure 1.15 The directed line segments $\vec{PQ}$ and $\vec{QP}$ in space.

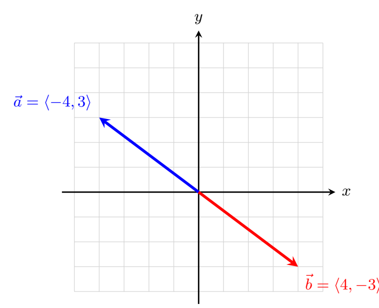

Figure 1.16 The corresponding position vectors $\vec{a}$ and $\vec{b}$ at the origin.

▲

Definition 1.17 Nonzero vectors $\vec{a}$ and $\vec{b}$ in V_2 have:

1 the **same direction** if $\vec{b} = c\vec{a}$ for some scalar $c > 0$

2 the **opposite direction** if $\vec{b} = c\vec{a}$ for some scalar $c < 0$

◆

Theorem 1.18 Magnitude of a Scalar Multiple. *If $\vec{a}$ is a vector and c is a scalar, then $\|c\vec{a}\| = |c|\|\vec{a}\|$.*

Definition 1.19 A vector of length 1 is called a **unit vector**. The **standard unit vectors** are:

$$\vec{i} = \langle 1, 0 \rangle, \quad \vec{j} = \langle 0, 1 \rangle$$

If $\vec{a} = \langle a_1, a_2 \rangle$, then it can be written as a linear combination of the standard unit vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j}$$

◆

Example 1.20 If $\vec{a} = -5\vec{i} + 2\vec{j}$ and $\vec{b} = \vec{i} - 3\vec{j}$, express $2\vec{a} - 3\vec{b}$ as a linear combination of $\vec{i}$ and $\vec{j}$.

Solution. Substitute the linear combinations for $\vec{a}$ and $\vec{b}$ into the expression, distribute the scalars, and combine like terms:

$$\begin{aligned} 2\vec{a} - 3\vec{b} &= 2(-5\vec{i} + 2\vec{j}) - 3(\vec{i} - 3\vec{j}) \\ &= (-10\vec{i} + 4\vec{j}) - (3\vec{i} - 9\vec{j}) \\ &= -10\vec{i} - 3\vec{i} + 4\vec{j} + 9\vec{j} \\ &= -13\vec{i} + 13\vec{j} \end{aligned}$$

▲

Theorem 1.21 Unit Vector in the Same Direction. *A* ***unit vector*** *is a vector of magnitude 1. The vectors $\vec{i}$ and $\vec{j}$ are unit vectors. If $\vec{a} \neq \vec{0}$, then the unit vector $\vec{u}$ that has the same direction as $\vec{a}$ is:*

$$\vec{u} = \frac{1}{\|\vec{a}\|}\vec{a}$$

Example 1.22 Find the unit vector $\vec{u}$ that has the same direction as $\vec{a} = -8\vec{i} + 15\vec{j}$.

Solution. First, find the magnitude of vector $\vec{a}$:

$$\|\vec{a}\| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$$

Next, multiply the reciprocal of the magnitude by the original vector $\vec{a}$:

$$\vec{u} = \frac{1}{17}(-8\vec{i} + 15\vec{j}) = -\frac{8}{17}\vec{i} + \frac{15}{17}\vec{j}$$

▲

Example 1.23 A quarterback releases the football with a velocity of 50 ft/sec at an angle of 35° with the horizontal. Approximate the horizontal and vertical components of the vector.

Solution. A vector $\vec{v}$ given a magnitude $\|\vec{v}\|$ and a direction angle θ can be written in component form as $\vec{v} = \langle \|\vec{v}\|\cos(\theta), \|\vec{v}\|\sin(\theta)\rangle$.

Given $\|\vec{v}\| = 50$ and $\theta = 35°$:

- Horizontal component: $50\cos(35°) \approx 40.96$ ft/sec
- Vertical component: $50\sin(35°) \approx 28.68$ ft/sec

In component form, the velocity vector is approximately $\langle 40.96, 28.68\rangle$. ▲

Example 1.24 A jet airplane approaches a runway at an angle of 7.5° with the horizontal, traveling at a velocity of 160 mi/hr. Approximate the horizontal and vertical components of the vector.

Solution. Given a magnitude of $\|\vec{v}\| = 160$ and an approach angle of $\theta = 7.5°$:

- Horizontal component: $160\cos(7.5°) \approx 158.63$ mi/hr
- Vertical component: $160\sin(7.5°) \approx 20.88$ mi/hr

In component form, the velocity vector is approximately $\langle 158.63, 20.88\rangle$. Note that because the plane is descending, the vertical velocity component can be interpreted as acting in the downward direction (-20.88). ▲

1.2 Vectors in Three Dimensions

In order to represent points in space, we first choose a fixed point O (the origin) and three directed lines through O which are perpendicular to each other, called the **coordinate axes**. They are labeled as the x**-axis**, y**-axis**, and z**-axis**. Think of the x- and y-axes as being horizontal while the z-axis is vertical. The direction of the z-axis is determined by the **right-hand rule**.

The three coordinate axes determine the three **coordinate planes**. The xy-plane contains the x- and y-axes, the yz-plane contains the y- and z-axes, and the xz-plane contains the x- and z-axes. These three coordinate planes divide space into eight parts, called **octants**. The **first octant** is determined by the positive axes.

We represent a point P by the ordered triple (a, b, c) of real numbers and we call a, b, and c the **coordinates** of P; here, a, b, and c are the x-, y-, and z-coordinates, respectively.

The point $P(a, b, c)$ can be visualized as a corner of a rectangular box. If we drop a perpendicular from P to the xy-plane, we get a point Q with coordinates $(a, b, 0)$ called the **projection** of P on the xy-plane.

The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is the set of all ordered triples of real numbers and is denoted by $\mathbb{R}^3$.

Example 1.25 What surfaces in $\mathbb{R}^3$ are represented by the following equations?

1 $z = 3$

2 $y = 5$

Solution.

1 The equation $z = 3$ represents the set of all points $\{(x, y, z) \mid z = 3\}$ in $\mathbb{R}^3$. Since x and y can be any real numbers, this forms a horizontal plane parallel to the xy-plane, located exactly 3 units above it.

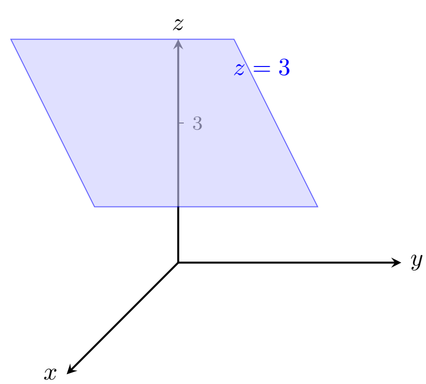

Figure 1.26 The plane $z = 3$ in $\mathbb{R}^3$.

2 The equation $y = 5$ represents the set of all points $\{(x, y, z) \mid y = 5\}$ in $\mathbb{R}^3$. Since x and z can be any real numbers, this forms a vertical plane parallel to the xz-plane, located exactly 5 units to the right of it.

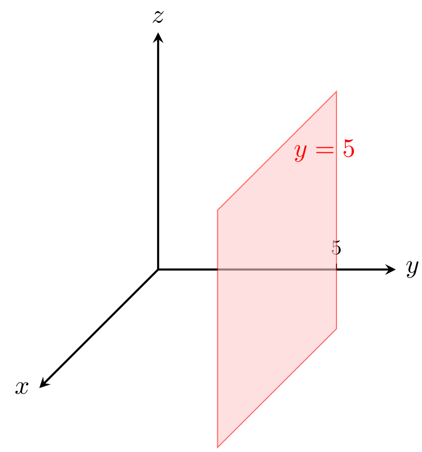

Figure 1.27 The plane $y = 5$ in $\mathbb{R}^3$.

▲

Fact 1.28 Distance Formula in Three Dimensions. *The distance* $d(P_1, P_2) = |P_1P_2|$ *between the points* $P_1(x_1, y_1, z_1)$ *and* $P_2(x_2, y_2, z_2)$ *is:*

$$d(P_1, P_2) = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 1.29 Find the distance from the point $P(-1, -3, 1)$ to the point $Q(3, 4, -2)$.

Solution. Substitute the coordinates of P and Q into the three-dimensional distance formula:

$$\begin{aligned} |PQ| &= \sqrt{(3 - (-1))^2 + (4 - (-3))^2 + (-2 - 1)^2} \\ &= \sqrt{(3 + 1)^2 + (4 + 3)^2 + (-3)^2} \\ &= \sqrt{4^2 + 7^2 + (-3)^2} \\ &= \sqrt{16 + 49 + 9} \\ &= \sqrt{74} \end{aligned}$$

The exact distance between the points is $\sqrt{74}$. ▲

Fact 1.30 Equation of a Sphere. *An equation of a sphere with center* $C(h, k, l)$ *and radius* r *is:*

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

If the center is at the origin $O(0, 0, 0)$*, then the equation of the sphere simplifies to:*

$$x^2 + y^2 + z^2 = r^2$$

Example 1.31 Show that $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ is the equation of a sphere, and find its center and radius.

Solution. To transform the given equation into standard form, group the variables and complete the square for each variable:

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) &= -6 \\ \left(x^2 + 4x + \left(\frac{4}{2}\right)^2\right) + \left(y^2 - 6y + \left(\frac{-6}{2}\right)^2\right) + \left(z^2 + 2z + \left(\frac{2}{2}\right)^2\right) &= -6 + 4 + 9 + 1 \\ (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) &= 8 \end{aligned}$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = 8$$

Comparing this result to the standard equation of a sphere $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$, we find:

- The center of the sphere is $C(h, k, l) = C(-2, 3, -1)$.
- The radius is $r = \sqrt{8} = 2\sqrt{2}$.

▲

Example 1.32 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$1 \leq x^2 + y^2 + z^2 \leq 4$$

Solution. The expression $x^2 + y^2 + z^2 = r^2$ represents the squared distance from the origin to a point in three-dimensional space. We can rewrite the inequality by taking the square root of all parts:

$$1 \leq \sqrt{x^2 + y^2 + z^2} \leq 2$$

This represents the set of all points whose distance r from the origin satisfies $1 \leq r \leq 2$. Therefore, the region is a solid **spherical shell** (or a thick hollow sphere) centered at the origin, including the inner sphere of radius 1 and the outer sphere of radius 2.

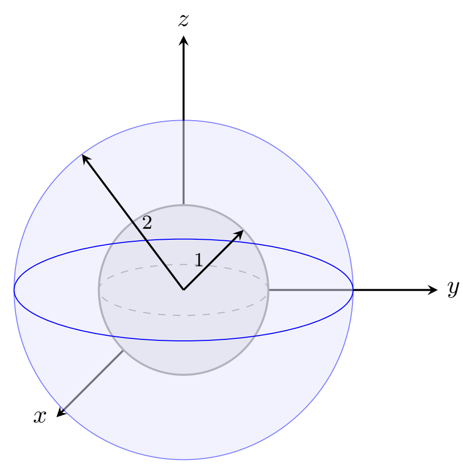

Figure 1.33 The spherical shell bounded by $r = 1$ and $r = 2$.

▲

Example 1.34 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$x^2 + y^2 \leq 25, \quad |z| \leq 3$$

Solution. We analyze the two conditions independently to find their intersection:

- In the xy-plane, $x^2 + y^2 \leq 25$ describes a solid disk of radius 5 centered at the origin. Since z is unconstrained by this expression, it forms an infinitely long solid cylinder extending along the z-axis.
- The absolute value inequality $|z| \leq 3$ is equivalent to $-3 \leq z \leq 3$. This bounds the space vertically between two horizontal planes.

Combining these conditions restricts the infinite cylinder to a specific height. Therefore, the region is a **solid vertical cylinder** of radius 5 and height 6, centered along the z-axis and extending from $z = -3$ to $z = 3$.

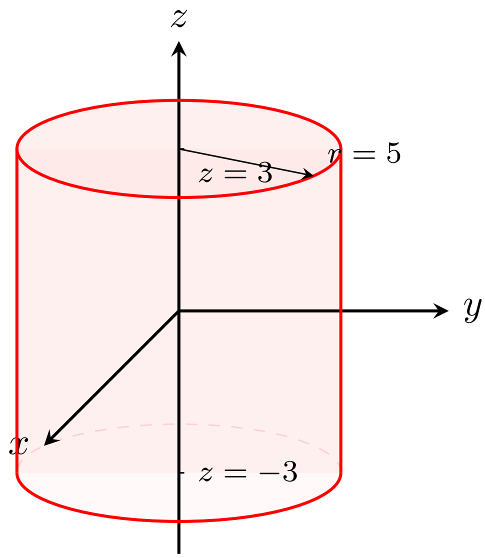

Figure 1.35 The solid vertical cylinder bounded by $x^2 + y^2 \leq 25$ and $-3 \leq z \leq 3$.

▲

Fact 1.36 Vector Operations in Three Dimensions. *Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ be vectors in $\mathbb{R}^3$.*

1 ***Addition of vectors:***

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

2 ***Scalar multiple of a vector:***

$$c\vec{a} = c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$

3 *The **zero vector 0**:*

$$\vec{0} = \langle 0, 0, 0 \rangle$$

4 *The **negative -a** of a vector $\vec{a}$:*

$$-\vec{a} = -\langle a_1, a_2, a_3 \rangle = \langle -a_1, -a_2, -a_3 \rangle$$

5 *The **difference a-b**:*

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

6 *The length of $\vec{a} = \langle a_1, a_2, a_3 \rangle$:*

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

7 *If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are any points, the vector $\vec{a}$ in $\mathbb{R}^3$ that corresponds to $P_1\vec{P}_2$ is:*

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Definition 1.37 The **standard basis vectors** in three dimensions are:

$$\vec{i} = \langle 1, 0, 0 \rangle, \quad \vec{j} = \langle 0, 1, 0 \rangle, \quad \vec{k} = \langle 0, 0, 1 \rangle$$

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$, then it can be written as a linear combination of the standard unit basis vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

◆

Example 1.38 If $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{b} = 4\vec{i} + 7\vec{k}$, express the vector $2\vec{a} + 3\vec{b}$ in terms of $\vec{i}$, $\vec{j}$, and $\vec{k}$.

Solution. Substitute the given expressions for $\vec{a}$ and $\vec{b}$, distribute the scalars, and group the components:

$$\begin{aligned} 2\vec{a} + 3\vec{b} &= 2(\vec{i} + 2\vec{j} - 3\vec{k}) + 3(4\vec{i} + 0\vec{j} + 7\vec{k}) \\ &= (2\vec{i} + 4\vec{j} - 6\vec{k}) + (12\vec{i} + 0\vec{j} + 21\vec{k}) \\ &= (2 + 12)\vec{i} + (4 + 0)\vec{j} + (-6 + 21)\vec{k} \\ &= 14\vec{i} + 4\vec{j} + 15\vec{k} \end{aligned}$$

▲

Example 1.39 Find the unit vector in the direction of the vector $\vec{v} = 2\vec{i} - \vec{j} - 2\vec{k}$.

Solution. First, find the magnitude of the given vector $\vec{v}$:

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

Next, multiply the reciprocal of the magnitude by the vector $\vec{v}$ to produce the unit vector $\vec{u}$:

$$\vec{u} = \frac{1}{3}(2\vec{i} - \vec{j} - 2\vec{k}) = \frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} - \frac{2}{3}\vec{k}$$

▲

Example 1.40 A 100-lb weight hangs from two wires making angles of 50° and 32° with the horizontal. Find the tensions (forces) $\vec{T}_1$ and $\vec{T}_2$ in both wires and their magnitudes.

Solution. We express the tension vectors in terms of their horizontal and vertical components. Let $\vec{T}_1$ pull up and to the left, and $\vec{T}_2$ pull up and to the right:

$$\begin{aligned} \vec{T}_1 &= -\|\vec{T}_1\|\cos(50°)\vec{i} + \|\vec{T}_1\|\sin(50°)\vec{j} \\ \vec{T}_2 &= \|\vec{T}_2\|\cos(32°)\vec{i} + \|\vec{T}_2\|\sin(32°)\vec{j} \end{aligned}$$

The downward force due to the weight is $\vec{w} = -100\vec{j}$. For the system to achieve static equilibrium, the resultant sum of the tensions must exactly counterbalance the weight:

$$\vec{T}_1 + \vec{T}_2 = -\vec{w} = 100\vec{j}$$

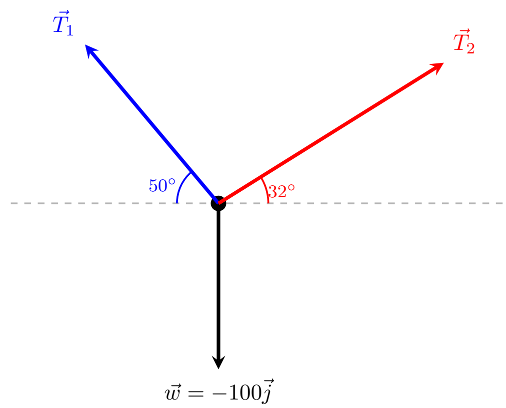

Figure 1.41 Force vector diagram for the hanging weight system.

Equating the horizontal ($\vec{i}$) and vertical ($\vec{j}$) components gives a system of linear equations:

$$-\|\vec{T}_1\|\cos(50^\circ) + \|\vec{T}_2\|\cos(32^\circ) = 0$$
$$\|\vec{T}_1\|\sin(50^\circ) + \|\vec{T}_2\|\sin(32^\circ) = 100$$

From the first equation, we find $\|\vec{T}_2\| = \|\vec{T}_1\|\frac{\cos(50^\circ)}{\cos(32^\circ)}$. Substituting this relationship into the second equation yields:

$$\|\vec{T}_1\|\sin(50^\circ) + \|\vec{T}_1\|\frac{\cos(50^\circ)\sin(32^\circ)}{\cos(32^\circ)} = 100$$

$$\|\vec{T}_1\|\left(\sin(50^\circ) + \cos(50^\circ)\tan(32^\circ)\right) = 100$$

Solving for the magnitudes yields:

- $\|\vec{T}_1\| \approx 85.64$ lb
- $\|\vec{T}_2\| = 85.64 \cdot \dfrac{\cos(50^\circ)}{\cos(32^\circ)} \approx 64.91$ lb

Substituting these scalar magnitudes back into our component formulas provides the final tension vector models:

$$\vec{T}_1 \approx -55.05\vec{i} + 65.60\vec{j}$$

$$\vec{T}_2 \approx 55.05\vec{i} + 34.40\vec{j}$$

▲

1.3 The Dot Product

Suppose a constant **force** is represented by a vector $\vec{F} = \vec{PR}$ pointing in some direction other than the direction of motion. If this force moves an object from point P to point Q, then the **displacement vector** is $\vec{D} = \vec{PQ}$.

The **work** done by $\vec{F}$ is defined as the magnitude of the displacement $\vec{D}$, multiplied by the magnitude of the applied force in the direction of the motion. Therefore, the work done by $\vec{F}$ is defined to be:

$$W = \|\vec{D}\|(\|\vec{F}\|\cos(\theta)) = \|\vec{D}\|\|\vec{F}\|\cos(\theta)$$

where θ is the angle between the force and displacement vectors.

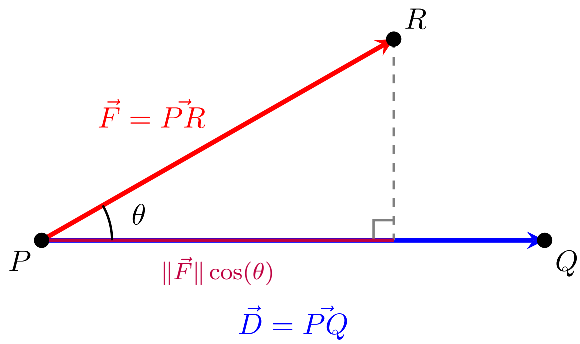

Figure 1.42 The geometric projection of a force vector $\vec{F}$ along a displacement vector $\vec{D}$.

Definition 1.43 The **dot product** of two vectors $\vec{a}$ and $\vec{b}$ is the number:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$$

where θ is the angle between $\vec{a}$ and $\vec{b}$, such that $0 \leq \theta \leq \pi$. Thus, θ is the smallest angle between the vectors when they are drawn with the same initial point. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, we define $\vec{a} \cdot \vec{b} = 0$. ◆

The dot product is a real number, that is, a scalar. For this reason, the dot product is sometimes called the **scalar product**.

Example 1.44 If the vectors $\vec{a}$ and $\vec{b}$ have lengths 4 and 6, and the angle between them is $\pi/3$, find $\vec{a} \cdot \vec{b}$.

Solution. Substitute the given magnitudes and the angle into the geometric definition of the dot product:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \|\vec{a}\|\|\vec{b}\|\cos(\theta) \\ &= (4)(6)\cos\left(\frac{\pi}{3}\right) \\ &= 24 \cdot \left(\frac{1}{2}\right) \\ &= 12 \end{aligned}$$

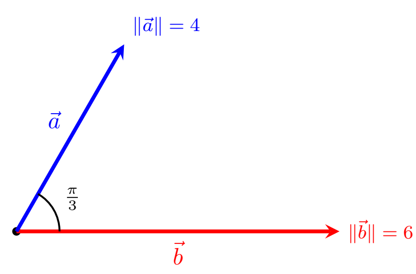

Figure 1.45 Two vectors separated by an angle of $\pi/3$.

▲

Example 1.46 A crate is hauled 8 m up a ramp under a constant force of 200 N applied at an angle of 25° to the ramp. Find the work done.

Solution. The work done by a constant force is given by the formula $W = \|\vec{F}\|\|\vec{D}\|\cos(\theta)$, which corresponds exactly to the dot product of the force vector and the displacement vector $\vec{F} \cdot \vec{D}$.

Given $\|\vec{F}\| = 200$ N, $\|\vec{D}\| = 8$ m, and $\theta = 25°$:

$$\begin{aligned} W &= (200)(8)\cos(25°) \\ &= 1600\cos(25°) \\ &\approx 1450.09 \text{ J} \end{aligned}$$

The work done in hauling the crate is approximately 1450.09 Joules (or Newton-meters).

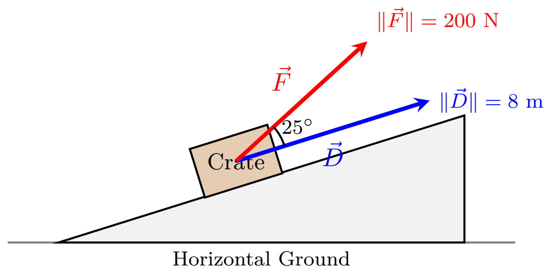

Figure 1.47 Force vector acting at an angle to the displacement along a ramp.

▲

Corollary 1.48 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are called **perpendicular** or **orthogonal** if the angle between them is $\theta = \pi/2$. In this case:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\pi/2) = 0$$

Conversely, if $\vec{a} \cdot \vec{b} = 0$ then $\cos(\theta) = 0$, which implies $\theta = \pi/2$. Therefore, two vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$.

Proof. By definition, the geometric dot product is given by $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$, where $0 \leq \theta \leq \pi$.

For the forward direction, assume $\vec{a}$ and $\vec{b}$ are orthogonal. By definition, the angle between them is $\theta = \pi/2$. Substituting this value yields:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\left(\frac{\pi}{2}\right) = \|\vec{a}\|\|\vec{b}\|(0) = 0$$

For the reverse direction, assume $\vec{a} \cdot \vec{b} = 0$ for two nonzero vectors. Since the vectors are nonzero, their magnitudes are strictly positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). We can divide both sides of the definition by these magnitudes:

$$0 = \|\vec{a}\|\|\vec{b}\|\cos(\theta) \implies \cos(\theta) = 0$$

Within the restricted interval $0 \leq \theta \leq \pi$, the equation $\cos(\theta) = 0$ has a unique solution, $\theta = \pi/2$. Thus, the vectors are orthogonal. ■

Corollary 1.49 *The sign of the dot product $\vec{a} \cdot \vec{b}$ provides geometric information about the relative directions of the vectors:*

1. *It is positive if $\vec{a}$ and $\vec{b}$ point in the same general direction ($0 \leq \theta < \pi/2$).*

2. *It is 0 if they are perpendicular ($\theta = \pi/2$).*

3. *It is negative if $\vec{a}$ and $\vec{b}$ point in generally opposite directions ($\pi/2 < \theta \leq \pi$).*

4. *If $\vec{a}$ and $\vec{b}$ point in exactly the same direction, then $\theta = 0$, which means*

$\cos(\theta) = 1$ *and:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|$$

5 *If* $\vec{a}$ *and* $\vec{b}$ *point in exactly the opposite direction, then* $\theta = \pi$*, which means* $\cos(\theta) = -1$ *and:*

$$\vec{a} \cdot \vec{b} = -\|\vec{a}\|\|\vec{b}\|$$

Proof. Because the magnitudes $\|\vec{a}\|$ and $\|\vec{b}\|$ of two nonzero vectors are always strictly positive, the sign of the product $\|\vec{a}\|\|\vec{b}\|\cos(\theta)$ is completely determined by the sign of the trigonometric term $\cos(\theta)$ on the interval $[0, \pi]$:

1 When $0 \leq \theta < \pi/2$ (an acute angle), $\cos(\theta) > 0$, rendering the dot product positive.

2 When $\theta = \pi/2$ (a right angle), $\cos(\theta) = 0$, rendering the dot product zero.

3 When $\pi/2 < \theta \leq \pi$ (an obtuse angle), $\cos(\theta) < 0$, rendering the dot product negative.

4 If they point in the exact same direction, $\theta = 0$. Since $\cos(0) = 1$, the expression simplifies directly to $\|\vec{a}\|\|\vec{b}\|$.

5 If they point in the exact opposite direction, $\theta = \pi$. Since $\cos(\pi) = -1$, the expression simplifies directly to $-\|\vec{a}\|\|\vec{b}\|$.

■

The Dot Product in Component Form

Suppose $\vec{a} = \langle a_1, a_2, a_3\rangle$ and $\vec{b} = \langle b_1, b_2, b_3\rangle$ are given in component forms. If we apply the Law of Cosines to the triangle formed by these vectors, we obtain:

$$\|\vec{a} - \vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\vec{a} \cdot \vec{b}$$

Fact 1.50 Algebraic Dot Product Formula. *The dot product of* $\vec{a} = \langle a_1, a_2, a_3\rangle$ *and* $\vec{b} = \langle b_1, b_2, b_3\rangle$ *is:*

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Example 1.51 Find the dot product $\vec{a} \cdot \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4\rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2}\rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Multiply corresponding components and add them together:

$$\vec{a} \cdot \vec{b} = (2)(6) + (-3)(2) + (4)\left(-\frac{1}{2}\right) = 12 - 6 - 2 = 4$$

2 First write $\vec{a}$ in full component form as $\langle 0, 4, -3\rangle$ and $\vec{b}$ as $\langle 2, 4, 6\rangle$:

$$\vec{a} \cdot \vec{b} = (0)(2) + (4)(4) + (-3)(6) = 0 + 16 - 18 = -2$$

3 Since components are unknown but the angle is provided, use the geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta) = (6)(5) \cos\left(\frac{2\pi}{3}\right) = 30\left(-\frac{1}{2}\right) = -15$$

▲

Example 1.52 Show that $2\vec{i} + 2\vec{j} - \vec{k}$ is perpendicular to $5\vec{i} - 4\vec{j} + 2\vec{k}$.

Solution. Two vectors are perpendicular (orthogonal) if and only if their dot product equals zero. Let $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -4, 2\rangle$:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-4) + (-1)(2) = 10 - 8 - 2 = 0$$

Because the dot product is exactly 0, the vectors are perpendicular. ▲

Example 1.53 Find the angle between the vectors $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -3, 2\rangle$.

Solution. We use the alternate formula for the angle between vectors: $\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}$.

First compute the dot product and individual vector magnitudes:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-3) + (-1)(2) = 10 - 6 - 2 = 2$$

$$\|\vec{a}\| = \sqrt{2^2 + 2^2 + (-1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$
$$\|\vec{b}\| = \sqrt{5^2 + (-3)^2 + 2^2} = \sqrt{25 + 9 + 4} = \sqrt{38}$$

Substitute these calculations back into the cosine expression:

$$\cos(\theta) = \frac{2}{3\sqrt{38}} \implies \theta = \arccos\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46 \text{ radians (or } 83.8^\circ)$$

▲

Proposition 1.54 Properties of the Dot Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$

2 $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

3 $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

4 $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$

5 $\vec{0} \cdot \vec{a} = \vec{a} \cdot \vec{0} = 0$

Proof. Let $\vec{a} = \langle a_1, a_2, a_3\rangle$, $\vec{b} = \langle b_1, b_2, b_3\rangle$, and $\vec{c} = \langle c_1, c_2, c_3\rangle$.

1 By the component formula for the dot product:

$$\vec{a} \cdot \vec{a} = a_1a_1 + a_2a_2 + a_3a_3 = a_1^2 + a_2^2 + a_3^2 = (\|\vec{a}\|)^2 = \|\vec{a}\|^2$$

2 Since multiplication of real numbers is commutative ($a_ib_i = b_ia_i$):

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = b_1a_1 + b_2a_2 + b_3a_3 = \vec{b} \cdot \vec{a}$$

3 Using the distributive property of real numbers:

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \langle b_1 + c_1, b_2 + c_2, b_3 + c_3\rangle$$

$$
\begin{aligned}
&= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\
&= (a_1b_1 + a_1c_1) + (a_2b_2 + a_2c_2) + (a_3b_3 + a_3c_3) \\
&= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}
\end{aligned}
$$

4 By factoring the real scalar c from the algebraic terms:

$$(c\vec{a}) \cdot \vec{b} = (ca_1)b_1 + (ca_2)b_2 + (ca_3)b_3 = c(a_1b_1 + a_2b_2 + a_3b_3) = c(\vec{a} \cdot \vec{b})$$

A symmetric argument shows this is also equal to $\vec{a} \cdot (c\vec{b})$.

5 Since multiplying any real number by zero equals zero:

$$\vec{0} \cdot \vec{a} = (0)(a_1) + (0)(a_2) + (0)(a_3) = 0 + 0 + 0 = 0$$

■

Theorem 1.55 Cauchy-Schwarz Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

Proof. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then both sides of the inequality are exactly 0, making the statement trivially true.

Assume both vectors are nonzero. We can express the dot product using its geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta)$$

Taking the absolute value of both sides yields:

$$|\vec{a} \cdot \vec{b}| = |\|\vec{a}\|\|\vec{b}\| \cos(\theta)| = \|\vec{a}\|\|\vec{b}\||\cos(\theta)|$$

Since the cosine function is naturally bounded by the range of real values $-1 \leq \cos(\theta) \leq 1$, its absolute value parameter satisfies $|\cos(\theta)| \leq 1$. Substituting this boundary restriction into our equation produces the target proof expression:

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|(1) \implies |\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

■

Theorem 1.56 Triangle Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|$$

Proof. Because both sides of the target inequality are non-negative real lengths, we can square the left-hand side and evaluate it using dot product properties:

$$
\begin{aligned}
\|\vec{a} + \vec{b}\|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\
&= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\
&= \|\vec{a}\|^2 + 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2
\end{aligned}
$$

Since a real number is always less than or equal to its absolute value ($x \leq |x|$), we can state that $\vec{a} \cdot \vec{b} \leq |\vec{a} \cdot \vec{b}|$. Applying the Cauchy-Schwarz Inequality ($|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$) gives:

$$
\begin{aligned}
\|\vec{a} + \vec{b}\|^2 &\leq \|\vec{a}\|^2 + 2|\vec{a} \cdot \vec{b}| + \|\vec{b}\|^2 \\
&\leq \|\vec{a}\|^2 + 2\|\vec{a}\|\|\vec{b}\| + \|\vec{b}\|^2
\end{aligned}
$$

The right side is a perfect square trinomial, which factors into scalar lengths:

$$\|\vec{a}+\vec{b}\|^2 \le (\|\vec{a}\|+\|\vec{b}\|)^2$$

Taking the positive square root of both sides completes the proof:

$$\|\vec{a}+\vec{b}\| \le \|\vec{a}\|+\|\vec{b}\|$$

■

Projections

Let $\vec{PQ}$ and $\vec{PR}$ be representations of two vectors $\vec{a}$ and $\vec{b}$ sharing the same initial point P. If S is the foot of the perpendicular dropped from R to the line containing $\vec{PQ}$, then the vector represented by $\vec{PS}$ is called the **vector projection** of $\vec{b}$ onto $\vec{a}$ and is denoted by $\operatorname{proj}_{\vec{a}} \vec{b}$ (think of this as the shadow cast by $\vec{b}$ onto $\vec{a}$).

The **scalar projection** of $\vec{b}$ onto $\vec{a}$ (also called the **component of b along a**) is the real number $\|\vec{b}\|\cos(\theta)$, where θ is the angle between $\vec{a}$ and $\vec{b}$. This value is denoted by $\operatorname{comp}_{\vec{a}} \vec{b}$. The equation:

$$\vec{a}\cdot\vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|(\|\vec{b}\|\cos(\theta))$$

shows that the dot product of $\vec{a}$ and $\vec{b}$ can be interpreted as the length of $\vec{a}$ multiplied by the scalar projection of $\vec{b}$ onto $\vec{a}$. Since:

$$\|\vec{b}\|\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{\vec{a}}{\|\vec{a}\|}\cdot\vec{b}$$

Fact 1.57 Projection Formulas. *Let $\vec{a}$ and $\vec{b}$ be vectors with $\vec{a} \neq \vec{0}$.*

- ***Scalar projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{comp}_{\vec{a}} \vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}$$

- ***Vector projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{proj}_{\vec{a}} \vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}\right)\frac{\vec{a}}{\|\vec{a}\|} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\vec{a}$$

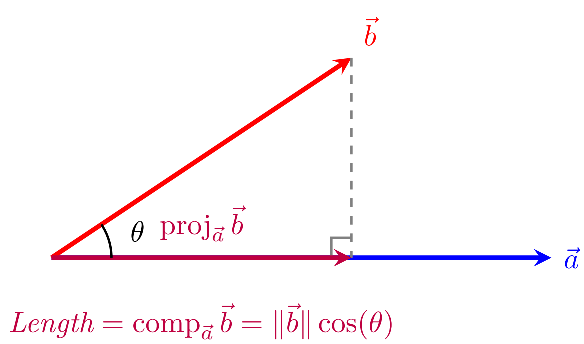

Figure 1.58 *The geometric relationship between vector projection and scalar projection.*

Example 1.59 Find the scalar projection and vector projection of $\vec{b} = \langle 1, 1, 2\rangle$ onto $\vec{a} = \langle -2, 3, 1\rangle$.

Solution. First, compute the core values required by the projection formulas ($\vec{a}\cdot\vec{b}$ and $\|\vec{a}\|^2$):

$$\vec{a}\cdot\vec{b} = (-2)(1) + (3)(1) + (1)(2) = -2 + 3 + 2 = 3$$

$$\|\vec{a}\|^2 = (-2)^2 + 3^2 + 1^2 = 4 + 9 + 1 = 14 \implies \|\vec{a}\| = \sqrt{14}$$

Now apply these numbers to the scalar projection formula:

$$\operatorname{comp}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{3}{\sqrt{14}}$$

Next, apply the values to the vector projection formula to scale the base direction vector $\vec{a}$:

$$\operatorname{proj}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\vec{a} = \frac{3}{14}\langle -2, 3, 1\rangle = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14}\right\rangle$$

▲

Checkpoint 1.60 Find the direction angle between the vector $\vec{a} = \langle 1, 2, 3\rangle$ and the x-axis.

Solution. The direction angle relative to the positive x-axis matches the angle between vector $\vec{a}$ and the standard basis unit vector $\vec{i} = \langle 1, 0, 0\rangle$:

$$\vec{a}\cdot\vec{i} = (1)(1) + (2)(0) + (3)(0) = 1$$

$$\|\vec{a}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$
$$\|\vec{i}\| = 1$$

Now compute the angle tracking parameter α:

$$\cos(\alpha) = \frac{\vec{a}\cdot\vec{i}}{\|\vec{a}\|\|\vec{i}\|} = \frac{1}{\sqrt{14}} \implies \alpha = \arccos\left(\frac{1}{\sqrt{14}}\right) \approx 74.5^\circ$$

Checkpoint 1.61 A force is given by a vector $\vec{F} = 3\vec{i} + 4\vec{j} + 5\vec{k}$ and moves a particle from the point $P(2, 1, 0)$ to the point $Q(4, 6, 2)$. Find the work done.

Solution. First, determine the displacement vector $\vec{D}$ from P to Q using terminal minus initial coordinates:

$$\vec{D} = \vec{PQ} = \langle 4 - 2, 6 - 1, 2 - 0\rangle = \langle 2, 5, 2\rangle$$

Work is computed as the dot product of the force vector $\vec{F} = \langle 3, 4, 5\rangle$ and this displacement vector $\vec{D}$:

$$W = \vec{F}\cdot\vec{D} = (3)(2) + (4)(5) + (5)(2) = 6 + 20 + 10 = 36$$

The total work done is 36 units.

Checkpoint 1.62 A cart weighing 100 lb is pushed up an incline that makes an angle of 30° with the horizontal. Find the work done against gravity in pushing the cart a distance of 80 feet.

Solution. The work done against gravity depends entirely on the vertical displacement of the cart. Alternatively, using our standard projection model, the gravity force vector points straight down with magnitude 100 lb: $\vec{F}_g = -100\vec{j}$.

If we align our coordinate frame with the horizontal ground, a displacement of 80 feet along a 30° incline yields:

$$\vec{D} = 80\cos(30^\circ)\vec{i} + 80\sin(30^\circ)\vec{j} = 80\left(\frac{\sqrt{3}}{2}\right)\vec{i} + 80\left(\frac{1}{2}\right)\vec{j} = 40\sqrt{3}\vec{i} + 40\vec{j}$$

The work done by gravity is $\vec{F}_g \cdot \vec{D} = (0)(40\sqrt{3}) + (-100)(40) = -4000$ ft-lb. The work done **against** gravity is the positive opposite of this value:

$$W = 4000 \text{ ft-lb}$$

Checkpoint 1.63 A wagon is pulled a distance of 100 m along a horizontal path by a constant force of 70 N. The handle of the wagon is held at an angle of 35° above the horizontal. Find the work done by the force.

Solution. Work is defined as the magnitude of the displacement vector multiplied by the scalar projection of the force vector along the direction of motion:

$$W = \|\vec{D}\|\left(\text{comp}_{\vec{D}}\,\vec{F}\right) = \|\vec{D}\|\left(\|\vec{F}\|\cos(\theta)\right)$$

Substitute the given values ($\|\vec{D}\| = 100$ m, $\|\vec{F}\| = 70$ N, and $\theta = 35^\circ$) into the work expression:

$$\begin{aligned} W &= (100)(70)\cos(35^\circ) \\ &= 7000\cos(35^\circ) \\ &\approx 5734.06 \text{ J} \end{aligned}$$

The total work done by the force pulling the wagon is approximately 5734.06 Joules.

1.4 The Cross Product (The vector product)

The cross product of two vectors in three-dimensional space is a vector that is perpendicular to both of the original vectors. It is defined as:

$$\vec{a} \times \vec{b} = \|\vec{a}\|\|\vec{b}\|\sin(\theta)\vec{n}$$

where θ is the angle between the vectors and $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Definition 1.64 If $\vec{a}$ and $\vec{b}$ are nonzero three-dimensional vectors, the **cross product** of $\vec{a}$ and $\vec{b}$ is the vector:

$$\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\|\sin(\theta))\vec{n}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ such that $0 \le \theta \le \pi$, and $\vec{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. The direction of $\vec{n}$ is determined by the **right-hand rule**: if the fingers of your right hand curl through the angle θ from $\vec{a}$ to $\vec{b}$, then your thumb points in the direction of $\vec{n}$.

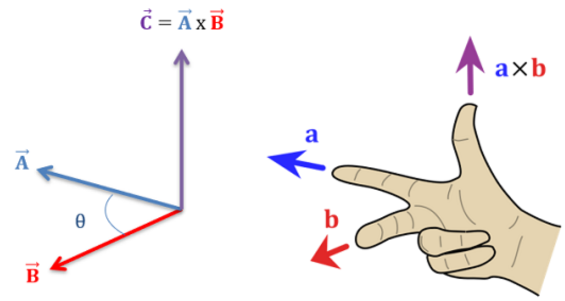

Figure 1.65 The right-hand rule and mutual vector orthogonality.

◆

Corollary 1.66 *The vector $\vec{a} \times \vec{b}$ points in the same direction as $\vec{n}$ because it is a scalar multiple of $\vec{n}$ with a non-negative scalar coefficient. Consequently, $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$. This is equivalent to stating:*

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0 \quad \textit{and} \quad (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

Proof. By definition, the cross product is a scalar multiple of the vector $\vec{n}$, meaning $\vec{a} \times \vec{b} = c\vec{n}$ where $c = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. Because $\vec{n}$ is explicitly defined as a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, its dot products with them must evaluate to zero:

$$\vec{n} \cdot \vec{a} = 0 \quad \text{and} \quad \vec{n} \cdot \vec{b} = 0$$

Using the scalar distribution properties of the dot product, we can substitute $c\vec{n}$ back into the orthogonality checks:

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (c\vec{n}) \cdot \vec{a} = c(\vec{n} \cdot \vec{a}) = c(0) = 0$$

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = (c\vec{n}) \cdot \vec{b} = c(\vec{n} \cdot \vec{b}) = c(0) = 0$$

This confirms that the resulting vector product remains perfectly perpendicular to both original structural components. ■

Corollary 1.67 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if the angle between them is either 0 or π. Therefore, two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if:*

$$\vec{a} \times \vec{b} = \vec{0}$$

Proof. The magnitude of the cross product vector is given by $\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. For any vector, its overall value equals the zero vector ($\vec{0}$) if and only if its absolute magnitude length scales to exactly 0.

Because we are evaluating two strictly nonzero vectors, their isolated scalar magnitudes are positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). Therefore, the product evaluates to zero if and only if the trigonometric term vanishes:

$$\|\vec{a} \times \vec{b}\| = 0 \iff \sin(\theta) = 0$$

On the restricted tracking interval $0 \leq \theta \leq \pi$, the condition $\sin(\theta) = 0$ is satisfied if and only if $\theta = 0$ or $\theta = \pi$. Geometrically, an angle of 0 means the vectors point in the exact same direction, and an angle of π means they point in exactly opposite directions. In either case, the vectors are parallel. ■

Example 1.68 Find $\vec{i} \times \vec{j}$ and $\vec{j} \times \vec{i}$.

Solution. We use the geometric definition of the cross product, $\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\| \sin(\theta))\vec{n}$:

- For $\vec{i} \times \vec{j}$, both vectors are unit vectors, so $\|\vec{i}\| = 1$ and $\|\vec{j}\| = 1$. The standard basis vectors are mutually perpendicular, meaning the angle between them is $\theta = \pi/2$. By the right-hand rule, curling your fingers from the positive x-axis ($\vec{i}$) to the positive y-axis ($\vec{j}$) forces your thumb to point along the positive z-axis ($\vec{k}$):

$$\vec{i} \times \vec{j} = (1 \cdot 1 \cdot \sin(\pi/2))\vec{k} = (1 \cdot 1 \cdot 1)\vec{k} = \vec{k}$$

- For $\vec{j} \times \vec{i}$, reversing the order of operations reverses the orientation of your hand gesture. The thumb now points down along the negative z-axis ($-\vec{k}$):

$$\vec{j} \times \vec{i} = (1 \cdot 1 \cdot \sin(\pi/2))(-\vec{k}) = -\vec{k}$$

▲

Remark 1.69 The cross product of any vector with itself is always the zero vector, because the angle between identical direction paths is $\theta = 0$, and $\sin(0) = 0$:

$$\vec{i} \times \vec{i} = \vec{0}, \quad \vec{j} \times \vec{j} = \vec{0}, \quad \vec{k} \times \vec{k} = \vec{0}$$

The Cross Product in Component Form

Fact 1.70 Algebraic Cross Product Formula. *The cross product of* $\vec{a} = \langle a_1, a_2, a_3 \rangle$ *and* $\vec{b} = \langle b_1, b_2, b_3 \rangle$ *is:*

$$\vec{a} \times \vec{b} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle$$

The expression for $\vec{a} \times \vec{b}$ is easier to remember in determinant form. The cross product of $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ can be expanded along its top row as:

$$\vec{a} \times \vec{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}$$

We compact this row-expansion memory tool by writing it as a formal 3×3 matrix determinant:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Proposition 1.71 Properties of the Cross Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ *(Anti-commutativity)*

2 $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b}) = \vec{a} \times (c\vec{b})$

3 $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$ *(Left distributivity)*

4 $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$ *(Right distributivity)*

Proof. Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$, and $\vec{c} = \langle c_1, c_2, c_3 \rangle$. By algebraic definition, the cross product component vector expansion is:

$$\vec{a} \times \vec{b} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle$$

1 Expanding the reverse cross product $\vec{b} \times \vec{a}$ yields:

$$\vec{b} \times \vec{a} = \langle b_2a_3 - b_3a_2,\, b_3a_1 - b_1a_3,\, b_1a_2 - b_2a_1 \rangle$$

Factoring out a negative -1 from each scalar entry directly flips the interior subtraction ordering:

$$\begin{aligned}\vec{b} \times \vec{a} &= \langle -(a_2b_3 - a_3b_2),\, -(a_3b_1 - a_1b_3),\, -(a_1b_2 - a_2b_1) \rangle \\ &= -\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle = -(\vec{a} \times \vec{b})\end{aligned}$$

Multiplying both sides by -1 satisfies the anti-commutative condition $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$.

2 Multiplying vector $\vec{a}$ by a scalar c updates our definition to $c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$. Substituting this scalar into the expansion gives:

$$\begin{aligned}(c\vec{a}) \times \vec{b} &= \langle (ca_2)b_3 - (ca_3)b_2,\, (ca_3)b_1 - (ca_1)b_3,\, (ca_1)b_2 - (ca_2)b_1 \rangle \\ &= \langle c(a_2b_3 - a_3b_2),\, c(a_3b_1 - a_1b_3),\, c(a_1b_2 - a_2b_1) \rangle \\ &= c\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle = c(\vec{a} \times \vec{b})\end{aligned}$$

A symmetric factoring loop shows this expression is identically equal to $\vec{a} \times (c\vec{b})$.

3 First, compile the vector addition term: $\vec{b} + \vec{c} = \langle b_1 + c_1, b_2 + c_2, b_3 + c_3 \rangle$. Crossing $\vec{a}$ with this combined vector yields:

$$\begin{aligned}\vec{a} \times (\vec{b} + \vec{c}) &= \langle a_2(b_3 + c_3) - a_3(b_2 + c_2),\, a_3(b_1 + c_1) - a_1(b_3 + c_3),\, a_1(b_2 + c_2) - a_2(b_1 + c_1) \rangle \\ &= \langle a_2b_3 + a_2c_3 - a_3b_2 - a_3c_2,\, a_3b_1 + a_3c_1 - a_1b_3 - a_1c_3,\, a_1b_2 + a_1c_2 - a_2b_1 - a_2c_1 \rangle\end{aligned}$$

Regrouping the entries splits the single vector back into a clean linear addition step:

$$\begin{aligned}&= \langle (a_2b_3 - a_3b_2) + (a_2c_3 - a_3c_2),\, (a_3b_1 - a_1b_3) + (a_3c_1 - a_1c_3),\, (a_1b_2 - a_2b_1) + (a_1c_2 - a_2c_1) \rangle \\ &= \langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle + \langle a_2c_3 - a_3c_2,\, a_3c_1 - a_1c_3,\, a_1c_2 - a_2c_1 \rangle \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c}\end{aligned}$$

4 Property 4 can be proven directly by using the anti-commutative law established in step 1, combined with the left distributive property from step 3:

$$\begin{aligned}(\vec{a} + \vec{b}) \times \vec{c} &= -(\vec{c} \times (\vec{a} + \vec{b})) \\ &= -(\vec{c} \times \vec{a} + \vec{c} \times \vec{b}) \\ &= -(\vec{c} \times \vec{a}) - (\vec{c} \times \vec{b}) \\ &= (\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c})\end{aligned}$$

This completes the verification of all four arithmetic identities.

■

If $\vec{a}$ and $\vec{b}$ are represented by directed line segments with the same initial point, they determine a parallelogram. This parallelogram has a base of length $\|\vec{a}\|$ and an altitude height of $\|\vec{b}\| \sin(\theta)$. Therefore, the total area A is:

$$A = \|\vec{a}\| \|\vec{b}\| \sin(\theta) = \|\vec{a} \times \vec{b}\|$$

Remark 1.72 The length of the cross product vector $\|\vec{a} \times \vec{b}\|$ is equal to the area of the parallelogram determined by the vectors $\vec{a}$ and $\vec{b}$.

Example 1.73 Find the cross product $\vec{a} \times \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4\rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2}\rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Set up and evaluate the 3×3 determinant:

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 4 \\ 6 & 2 & -\frac{1}{2} \end{vmatrix} \\ &= \begin{vmatrix} -3 & 4 \\ 2 & -\frac{1}{2} \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 4 \\ 6 & -\frac{1}{2} \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & -3 \\ 6 & 2 \end{vmatrix} \vec{k} \\ &= \left(\frac{3}{2} - 8\right)\vec{i} - (-1 - 24)\vec{j} + (4 - (-18))\vec{k} \\ &= -\frac{13}{2}\vec{i} + 25\vec{j} + 22\vec{k} = \left\langle -\frac{13}{2}, 25, 22 \right\rangle\end{aligned}$$

2 Write $\vec{a}$ in full component form as $\langle 0, 4, -3\rangle$:

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 4 & -3 \\ 2 & 4 & 6 \end{vmatrix} \\ &= (24 - (-12))\vec{i} - (0 - (-6))\vec{j} + (0 - 8)\vec{k} \\ &= 36\vec{i} - 6\vec{j} - 8\vec{k} = \langle 36, -6, -8\rangle\end{aligned}$$

3 Because individual components are missing but the angle is given, we find the magnitude using the geometric formula and label the unit normal directional vector $\vec{n}$:

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta) = (6)(5) \sin\left(\frac{2\pi}{3}\right) = 30\left(\frac{\sqrt{3}}{2}\right) = 15\sqrt{3}$$

Thus, the cross product vector is $15\sqrt{3}\vec{n}$.

▲

Example 1.74 Find a vector perpendicular to the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. First, construct two independent vectors lying inside the plane using the given vertices:

$$\vec{PQ} = \langle -2 - 1, 5 - 4, -1 - 6\rangle = \langle -3, 1, -7\rangle$$

$$\vec{PR} = \langle 1 - 1, -1 - 4, 1 - 6\rangle = \langle 0, -5, -5\rangle$$

A vector perpendicular to the plane must be orthogonal to both paths. Compute their cross product:

$$\begin{aligned}\vec{PQ} \times \vec{PR} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} \\ &= (-5-35)\vec{i} - (15-0)\vec{j} + (15-0)\vec{k} \\ &= -40\vec{i} - 15\vec{j} + 15\vec{k} = \langle -40, -15, 15 \rangle\end{aligned}$$

Any non-zero scalar multiple of this vector (such as $\langle 8, 3, -3 \rangle$ obtained by dividing by -5) is also perpendicular to the plane. ▲

Example 1.75 Find the area of the triangle with vertices $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. The area of a triangle is exactly half the area of the parallelogram spanned by vectors sharing vertex P. From the previous example, we know $\vec{PQ} \times \vec{PR} = \langle -40, -15, 15 \rangle$.

Compute the magnitude length of this cross product vector:

$$\begin{aligned}\|\vec{PQ} \times \vec{PR}\| &= \sqrt{(-40)^2 + (-15)^2 + 15^2} \\ &= \sqrt{1600 + 225 + 225} = \sqrt{2050} = 5\sqrt{82}\end{aligned}$$

Therefore, the area of the triangle is:

$$\text{Area} = \frac{1}{2}\|\vec{PQ} \times \vec{PR}\| = \frac{5\sqrt{82}}{2} \approx 22.64$$

▲

Definition 1.76 The product $\vec{a} \cdot (\vec{b} \times \vec{c})$ is called the **scalar triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. ◆

The area of the base parallelogram is $A = \|\vec{b} \times \vec{c}\|$. If θ is the angle between vector $\vec{a}$ and the cross product vector $\vec{b} \times \vec{c}$, then the slanted vertical height h of the surrounding three-dimensional parallelepiped is $h = \|\vec{a}\||\cos(\theta)|$.

Fact 1.77 Volume of a Parallelepiped. *The volume of the parallelepiped determined by the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is the absolute magnitude of their scalar triple product:*

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

We can compute this product directly using a single matrix determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example 1.78 Use the scalar triple product to show that the vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$, and $\vec{c} = \langle 0, -9, 18 \rangle$ are coplanar, meaning they lie in the same plane.

Solution. Three vectors are coplanar if and only if the volume of the parallelepiped they span is exactly zero ($V = 0$). We check this condition by computing their scalar triple product determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}$$

$$
\begin{aligned}
&= 1\begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix} + (-7)\begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix} \\
&= 1(-18-(-36)) - 4(36-0) - 7(-18-0) \\
&= 1(18) - 4(36) - 7(-18) \\
&= 18 - 144 + 126 = 0
\end{aligned}
$$

Because the scalar triple product is exactly 0, the bounded volume vanishes. This proves that the three vectors are coplanar. ▲

Definition 1.79 The product $\vec{a} \times (\vec{b} \times \vec{c})$ is called the **vector triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. It can be expanded and computed using dot products via the BAC-CAB identity:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

◆

Example 1.80 Given the vectors $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}$, $\vec{b} = 2\vec{i} + \vec{k}$, and $\vec{c} = \vec{j} - 3\vec{k}$, evaluate the vector triple product $\vec{a} \times (\vec{b} \times \vec{c})$ using the expansion identity.

Solution. First, express the vectors cleanly in their numerical component forms:

$$\vec{a} = \langle 1, -1, 2 \rangle, \quad \vec{b} = \langle 2, 0, 1 \rangle, \quad \vec{c} = \langle 0, 1, -3 \rangle$$

Next, calculate the two scalar dot products required by the identity ($\vec{a} \cdot \vec{c}$ and $\vec{a} \cdot \vec{b}$):

$$
\begin{aligned}
\vec{a} \cdot \vec{c} &= (1)(0) + (-1)(1) + (2)(-3) = 0 - 1 - 6 = -7 \\
\vec{a} \cdot \vec{b} &= (1)(2) + (-1)(0) + (2)(1) = 2 + 0 + 2 = 4
\end{aligned}
$$

Substitute these scalar values back into the vector triple product identity formula:

$$
\begin{aligned}
\vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \\
&= -7\langle 2, 0, 1 \rangle - 4\langle 0, 1, -3 \rangle \\
&= \langle -14, 0, -7 \rangle - \langle 0, 4, -12 \rangle \\
&= \langle -14 - 0,\ 0 - 4,\ -7 - (-12) \rangle = \langle -14, -4, 5 \rangle
\end{aligned}
$$

In standard basis notation, the final vector is $-14\vec{i} - 4\vec{j} + 5\vec{k}$. ▲

Chapter 2

MyOpenMath Interactive Activities

These files contain copies of the interactive MyOpenMath exercises that guided our learning for each topic. Special thanks to Professor Jason Hardin for sharing his MyOpenMath question bank.

Chapter 3

Handouts

Chapter 4

Homework

Chapter 5

Course Documents

MA-105-BL: Survey of Math

Worcester State University, DGCE, Summer 2026

Course Information

Instructor	Dr. Hansun To, Professor of Mathematics, hto1@worcester.edu.
Meeting Information	This is a **Hybrid Course**, combining online and in-person learning to provide a flexible and tailored experience. You'll have the opportunity to learn at your own pace using MyOpenMath, participate in Zoom discussions as needed, and attend in-person sessions for exams. *In-person attendance is required* for the following sessions: • Two midterm exams (1 hour and 30 minutes): ◦ (In person) Thursday, June 4th, from 03:30 PM to 05:00 PM ◦ (Online) Thursday, June 25th. (The exam will be available on Thursday, June 25th, at 08:00 AM. Please submit your work by 06:30 PM, including any necessary show work attachments.) • Final exam (three hours): ◦ (In person) Thursday, July 2nd, from 03:30 PM to 05:30 PM All in-person exams will be held in Sullivan room S-136. Please plan accordingly to ensure your attendance at these mandatory sessions.
Prerequisite	Math placement exam code 3 or higher, or a weighted high school GPA of 2.7 or higher within the past 3 years

Responsibility for Learning. Every student is responsible for their own learning. The instructor will make every effort to support you; however, if you need any help it is your responsibility to reach out with questions

Course Content. In this course we will learn to improve the level of quantitative awareness of students using familiar situations that provide a sense of purpose for studying mathematics. The objective is not to make mathematicians of the students, but to help them deal as comfortably as possible with an environment that increasingly makes use of quantitative reasoning. Topics will include voting and apportionment methods, personal finance, set theory, and probability theory.

Required Materials. A calculator is required for this course. There is no required textbook; notes will be provided in class (or in pdf format) and other materials will be freely available on the MyOpenMath platform.

Calculators. A scientific, non-graphing calculator capable of computing logarithmic and trigonometric values is required, though its use will be limited to only a handful of sections. I recommend the ***TI-30X IIS***, which you can usually find for around $15. Graphing calculators, phones, tablets and any unauthorized devices are never allowed!

MyOpenMath. MyOpenMath (MOM) is a free, open source, online course management system designed for mathematics which we will use extensively in this course. It is available at https://www.myopenmath.com/ . You will need to create an account and register for our course on MyOpenMath to get started; we will do this as a group during the first class meeting. All course documents and materials will be posted on MyOpenMath; and all homework will be completed through MyOpenMath.

Assignments and Grading

Homework. Practice is a primary component of the mathematical learning process. For this reason, homework sets will be assigned on MyOpenMath after most class meetings. Assignments will always be due on Wednesdays by 11:59pm. This means you will often have two homework sets due on Wednesday, one which opened on the previous Monday and the other which opened on the previous Wednesday. While the number of problems in each homework set will vary, your score will be considered as a percentage and each set will be weighted equally when computing your homework grade. The problems on the homework sets are sometimes meant to deepen or extend the understanding you have gained from the class lecture and discussion; you should expect to find some of the exercises difficult.

The online homework is set up so you can try most problems two times, then the correct answer will be shown (there may be some questions in which you have only one try, this will be indicated in the homework set). If you miss a problem and want to improve your score, you can click the ``Try a similar problem'' link, which will give you a new question of the same type. You can keep on working on versions of a question until you get it correct. ***You must earn at least 50% on an assignment for your score to count towards your grade; any set in which you earn less than 50% will be counted as a 0.*** Assignments will remain available in "Practice" mode after their due date to provide additional practice of concepts; new versions of problems can be generated and answered, but scores will not be saved. ***Be aware, opening an online homework assignment in "Practice" mode will disqualify***

you from being able to redeem a LatePass on that assignment! If you are having trouble on a problem, you may use the "Message Instructor" link, which will allow you to send me a message through MyOpenMath; the specific problem you are working on will be automatically included in the message. I can then reply to the message with some guidance. These messages are internal to MyOpenMath and not email, so my reply will go to your messages when you login to MyOpenMath. I will check my messages on MyOpenMath at least once per day. Some problems may have a link to a video solution of a similar problem.

To account for any unexpected emergencies or technical issues that may occur, we will utilize the LatePass feature in MyOpenMath. A LatePass can be redeemed for an automatic 48-hour extension on any online homework set which has NOT been viewed in "Practice" mode. To use a LatePass, click the "Use LatePass" button next to the appropriate assignment on MyOpenMath and the platform will automatically extend the due date for that assignment to 48 hours after the original due date. Each student will be given 8 LatePasses to use as they wish throughout the semester. Note a maximum of two LatePasses can be used on any single assignment. If circumstances require a more substantial extension, contact your instructor as soon as possible to discuss this possibility.

Exams. There will be four exams in this course, one of which will be a cumulative final exam. Tentative dates are June 4, June 25, May 6, and July 2.

Grade Allocation. Your course grade will be determined by the following distribution. ***You must get at least a 50% on the final exam in order to pass the class***.

Assignment	**Percentage**
MyOpenMath Online Homework	10%
MyOpenMath Online Interactive Activities	10%
MyOpenMath Online Quizzes	10%
Exam 1	20%
Exam 2 (online)	20%
Final Exam	30%

Grading Scale. Letter grades will be assigned as follows. Grades will not be curved.

Letter Grade	**Percent Needed**
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
D+	67%
D	63%
D-	60%
E	Below 60%

Course Policies

Email. I try to respond to emails and MyOpenMath messages within 24 hours. You are encouraged to email/message regarding the homework; however, emails sent within 24 hours of a due date may not get a response before the deadline. So do not wait until the last minute to attempt the assignments!

Studying. To succeed in this course, you will need to do more than simply attend class. Mathematics is a subject which is learned by doing, meaning it is important to work problems outside of class to fully understand the concepts we cover. The assigned homework is the minimum amount you should be completing; if you feel like you don't fully understand a concept, ask me to guide you to some additional practice problems. A general rule for studying (in all college courses) is for each hour of class per week you spend a minimum of two hours outside of class on homework, studying, reading the text, etc.

Since this is a hybrid course and it is three credit hours, you should plan to spend at least six hours per week working on course material for a regular semester. Since this is a 7-week summer course, that time is doubled to 12 hours per week.

Academic Integrity and AI Usage. The WSU Mathematics Department does not tolerate academic dishonesty! For an explanation of what is considered "academic dishonesty" at WSU, please see the policy included in the 2025-2026 Undergraduate Catalog. All instructors in the WSU Mathematics Department are required to report all incidents of academic dishonesty observed, detected, or otherwise determined to have occurred to the WSU Academic Central File. WSU requires some penalty be imposed for every reported violation of the academic honesty policy. All incidents of academic dishonesty in this course will result in (at least) a score of 0 on that assignment; additional penalties will be imposed at the instructor's discretion and/or as required by the Mathematics Department Academic Honesty Policy.

Artificial Intelligence (AI) Policy: Artificial intelligence (AI) tools (ChatGPT and similar platforms) may be used for *study support and practice only.* Acceptable uses include reviewing concepts, generating practice problems, checking understanding of definitions, or exploring alternative explanations of course material. AI tools are not permitted on any graded assignments, quizzes, or examinations unless explicitly authorized by the instructor. This includes, but is not limited to, using AI to generate solutions, explanations, written work, computations, graphs, or code that is submitted for credit. All submitted work must reflect the student's own understanding and effort. Reliance on AI-generated responses in place of independent problem-solving undermines the learning objectives of this course and is not allowed. Students are responsible for ensuring that any work they submit complies with this policy. Use of AI in violation of this policy will be treated as a violation of the Academic Honesty Policy and may result in penalties including a score of zero on the assignment, quiz, or exam, and additional disciplinary action as required by university policy.

Instructor Rights. ***As the instructor of this course, I reserve the right to make changes to the above course policies at any time. All changes will be announced in class.***

Student Learning Outcomes

Students will be able to:

1. Students will be able to communicate their mathematical reasoning.
2. Financial Management: Students will be able to calculate simple and compound interest, loan installments and mortgage payments.
3. Probability Theory: Students will be able to determine sample spaces for experiments, calculate probabilities with fundamental counting principles, permutations and combinations.
4. Voting Methods: Students will be able to discuss the pros and cons of several voting methods and how they can be manipulated by a knowledgeable individual.

LASC

This is a Quantitative Reasoning (QR) course in the Liberal Arts and Sciences Curriculum (LASC).

The course addresses the following LASC Quantitative Reasoning objectives:

1. Acquaint students with formal systems, procedures, and sequences of operations.
2. Strengthen students' understanding of variables and functions.
3. Apply mathematical techniques to the analysis and solution of real-life problems.
4. Emphasize the importance of accuracy, including precise language and careful definitions of mathematical concepts.
5. Understand both the underlying principles and practical applications of one or more fields of mathematics.
6. Develop an understanding of and facility with statistical analysis an analysis, including understanding of its applications and limitations.

and the following LASC overarching objectives:

1. Communicate effectively orally and in writing.
2. Understand and employ quantitative and qualitative reasoning.
3. Apply skills in critical thinking.
4. Understand the roles of science and technology in our modern world.
5. Become socially responsible agents in the world.
6. Make connections across courses and disciplines.
7. Develop as healthy individuals--physically, emotionally, socially, ethically, and intellectually.

Course Schedule

Here is a tentative schedule of topics. We will try to stay on schedule as much as possible. If class is canceled for any reason, then the schedule will be adjusted to accommodate this cancelation. This could include a rearrangement of topics, removing a "review" day, or skipping a topic entirely, depending on the timing of the cancelation. Any changes to this schedule will be announced in class and via email.

Table 5.1 Course Schedule and Exam Dates

Unit	Topics	Due Dates
1	Voting methods, apportionment	Fri 5/22
2	Simple interest, compound interest	Wed 5/27
3	Annuities, installment loans ***(Quiz 1)***	Wed 5/27
4	Mortgages, credit cards	Wed 6/3
	In-Person Exam 1 Thu 6/4 (3:30-5:00pm) (S-136)	
5	Set basics, subsets	Wed 6/10
6	Set operations, surveys ***(Quiz 2)***	Wed 6/10
7	Intro to probability, events involving “not”, “or”, “and”	Wed 6/17
8	Counting, permutations, combinations ***(Quiz 3)***	Wed 6/17
	Online Exam 2 (Thu 6/25)	
9	Probability with counting, odds, conditional	Fri 6/26
	In-Person FINAL EXAM Thu 7/2 (3:30-6:30pm) (S-136)	